MUHAMMAD TALHA

AI / Machine Learning Engineer

+92 318 5026530 | talha987987@gmail.com | github.com/MuhammadTalha752 | linkedin.com/in/muhammad-talha-765b213a4 | Kahuta, Rawalpindi

Professional Summary

AI/ML-focused Computer Science graduate currently working as an AI Intern at APPINSNAP, where I build and evaluate production-style AI systems. My work covers LLM-powered customer-support agents with hybrid retrieval-augmented generation (RAG), computer-vision pipelines for document information extraction, a Transformer implemented from scratch in PyTorch, and research on Urdu/English speech models. I enjoy taking an idea from research and model comparison through to a working, testable application. Strong in Python, with additional experience in Java, C++, JavaScript, and cross-platform mobile development with React Native. Looking for AI/ML engineering roles where I can build practical, real-world AI products.

Technical Skills

AI / ML	Transformers, Self-Attention, NLP, LLM Agents, Prompt Engineering, RAG, Hybrid Search (BM25 + Dense), Reciprocal Rank Fusion
Frameworks & Tools	PyTorch, LangChain, Sentence-Transformers, YOLO, EasyOCR, OpenCV, Scikit-learn, Qdrant, Groq, Git, GitHub
Languages & DBs	Python, Java, C++, JavaScript, SQL, SQLite, MySQL

Experience

AI / Machine Learning Intern *July 2026 – Present*
APPINSNAP

- Designed and built LLM-based customer-support agents, first as a manual implementation (Groq, Llama 3.1 8B) and then with LangChain and hybrid RAG (Qdrant + BM25 with Reciprocal Rank Fusion), including user/session-based conversation memory.
- Developed an end-to-end Pakistani CNIC detection and information-extraction pipeline using YOLO, perspective correction, and EasyOCR, and tested it against non-CNIC documents to reduce false acceptances.
- Implemented a Transformer architecture from scratch in PyTorch to understand attention-based models at a low level, and trained it for text generation.
- Researched and benchmarked open-source Urdu and English TTS models (Chatterbox, Indic Parler-TTS) on pronunciation of Pakistani names, finance/app terminology, prosody, speed, and local GPU compatibility.
- Documented experiments and results and compared model options to support practical technical decisions.

React Native Mobile App Developer (Intern) *June 2025 – July 2025*
United Sol

- Built cross-platform mobile app screens and reusable UI components using React Native and JavaScript.
- Worked with a team using Git for version control and followed feedback from code reviews to improve code quality.

Selected AI Projects

Customer Support AI Agent (Manual & LangChain RAG) | *Python, LangChain, Qdrant, BM25, Groq, SQLite*
GitHub

- Built the agent twice: a framework-free version with SQLite and intent handling (register/check complaints, FAQ, unknown queries), then a LangChain version using `gpt-oss-120b`.
- Hybrid retrieval with all-MiniLM-L6-v2 embeddings in Qdrant plus BM25, merged using Reciprocal Rank Fusion, with session-based conversation persistence.

CNIC Detection & Information Extraction | *YOLO, OpenCV, EasyOCR, Python*

- YOLO-based detection and cropping, perspective correction, and OCR to extract name, father's name, CNIC number, gender, country, dates, and signature.
- Negative-image testing (cards, receipts, newspapers, other documents) to ensure non-CNIC inputs are rejected.

KidneyCare – CKD Detection System (Final Year Project) | *Java, Android, Machine Learning*

- Developed an Android app integrated with an ML model that predicts Chronic Kidney Disease from patient data, with data preprocessing and a mobile interface for real-time health insights.

More projects (Transformer from scratch, TTS research, and others): github.com/MuhammadTalha752

Education

BS Computer Science *2022 – 2026*
KICSIT (IST Sub-Campus) GPA: 3.24 / 4.00

Achievements & Languages

2nd position, Graphic Designing Competition, Friend's Vision Academy Kahuta (2022)
Languages: English (Fluent) | Urdu (Fluent) | Punjabi (Fluent)